THAMESIDE ERF (TBY) · PIPING ISOMETRIC · RECONSTRUCTED 3D / 2D PACKAGE

1LCN30 · DRG 2

DN40 condensate from DRG 1 through valves 1LCN30AA400 and 1LCN30AA030, down 645 mm into a horizontal expansion zig-zag at EL +4450, to the tie-in on 0LCN50BR200 (X 91628 · Y 82209 · EL +4450). Three bent-pipe spools, all bends 90° R 200.

Spool 1
6 032 mm
FW-1 → FW-2 · 2 × 90° R200

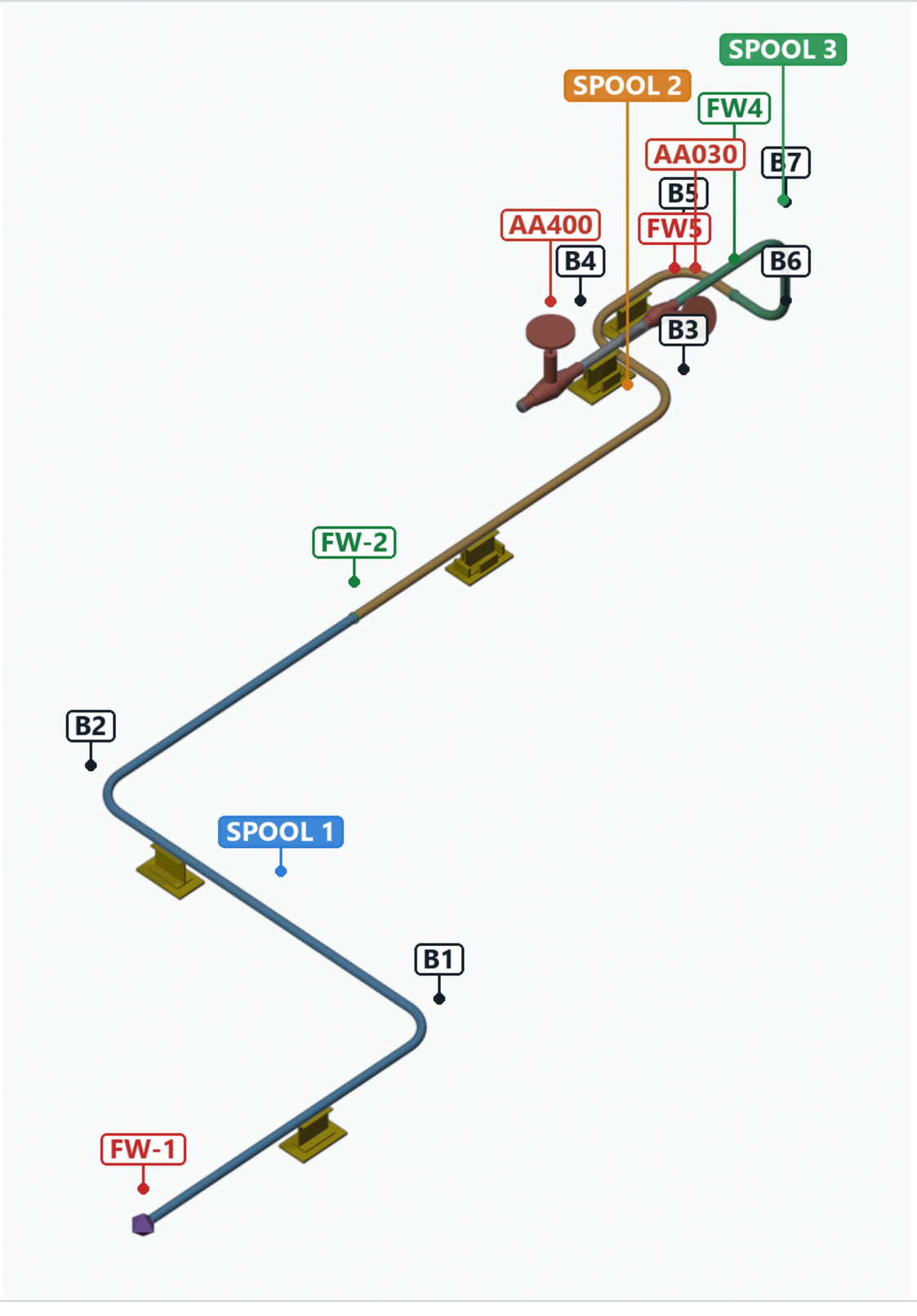
Spool 2
3 742 mm
FW-2 → FW4 · 3 × 90° R200

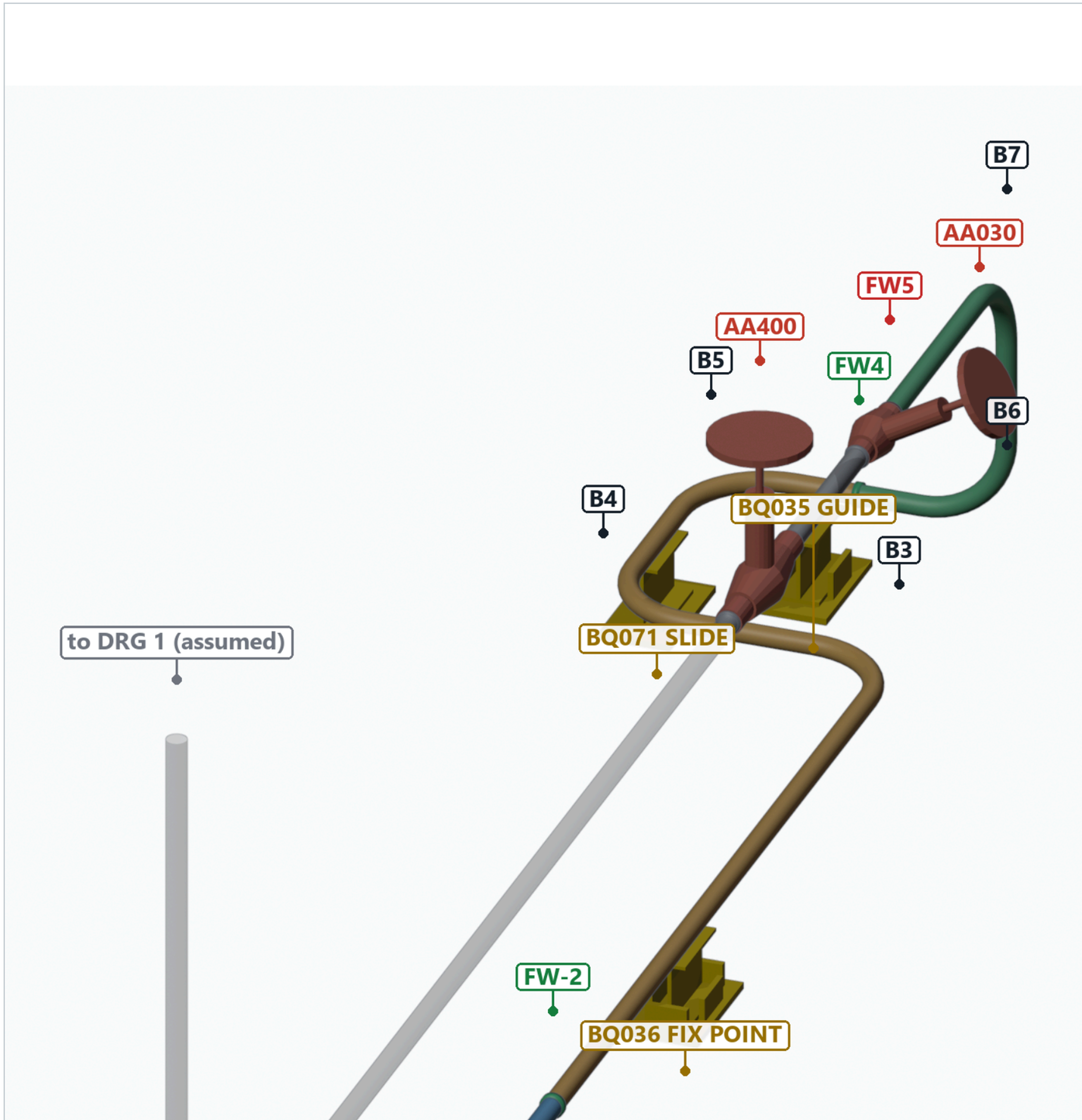
Spool 3
1 583 mm
FW4 → FW5 · 2 × 90° R200

Field welds
2 / 4 welded
FW-2 03.11.2026 · FW4 05.11.2026 · HC130

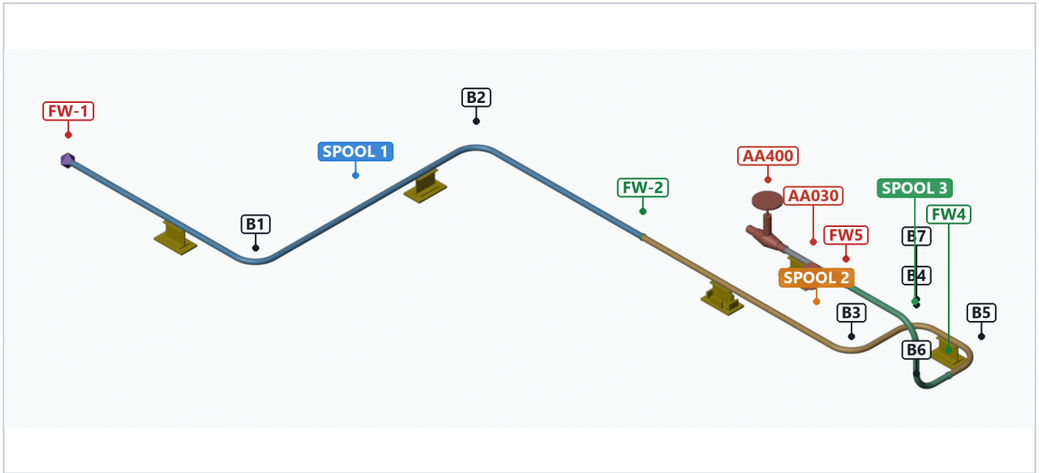
Medium	Condensate
Pipe class	ECJB_TBY
Nominal bore	DN40
Insulation thickness	100 mm
Insulation density	0 kg/m3
Fluid group / module	2 / Module A
PED category	Category I
Design pressure	96 bar(g)
Design temperature	309 deg C
Operating pressure	79.00 bar(g)
Operating temperature	295 deg C
Test pressure	PT08.01 - 207.95 bar(g)
P&ID drawing	90207086
Tie-in	0LCN50BR200 at X 91628 / Y 82209 / EL +4450
Upstream	CONT. FROM DRG 1
Bend radius	200 mm (all bends 90 deg)

Built from the site copy of the isometric with hand mark-up. Dimensions are theoretical PI-to-PI values; check on site before cutting (iso general remarks). Valve and support shapes are symbolic.

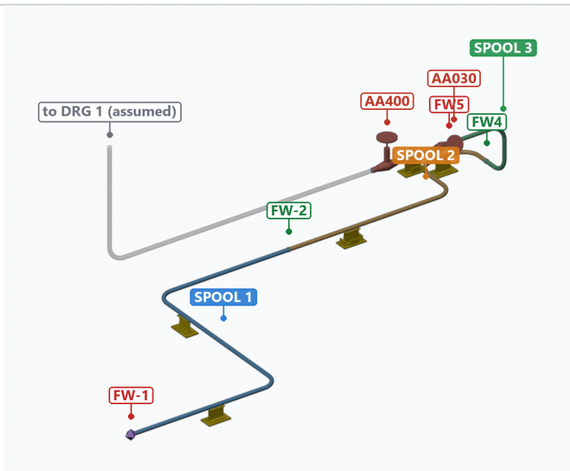




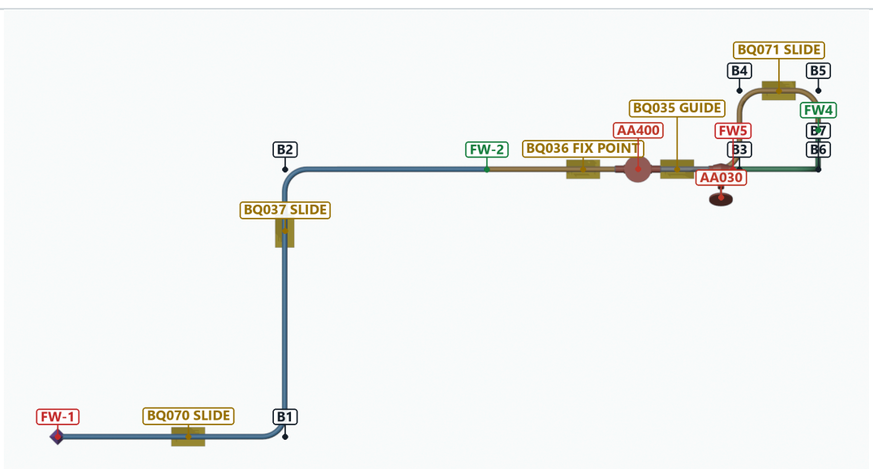
Valve run at EL +5095 and the U-loop, seen from the south-south-west



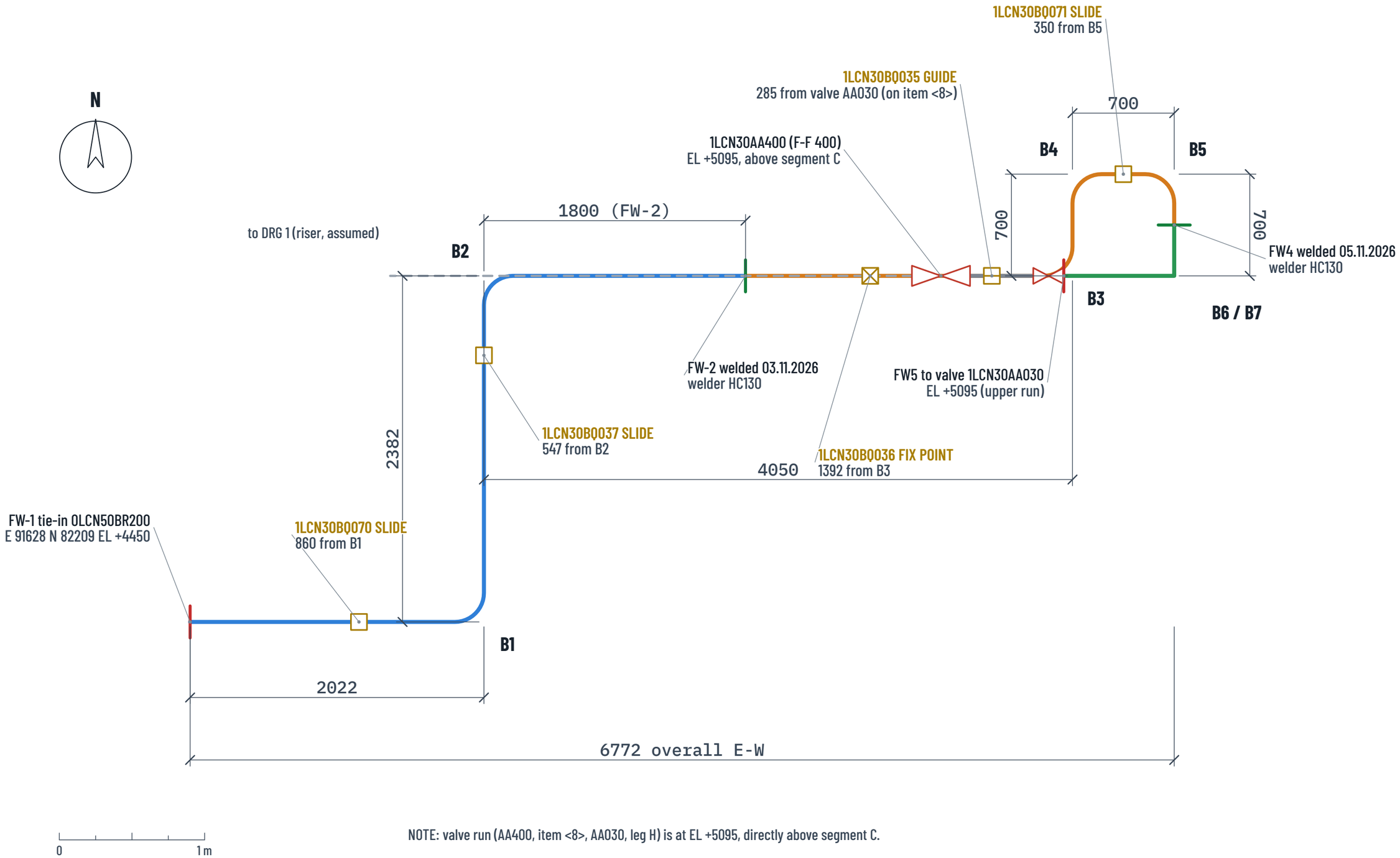
View from the south-east



With the DRG 1 continuation (grey, assumed riser)



Top view

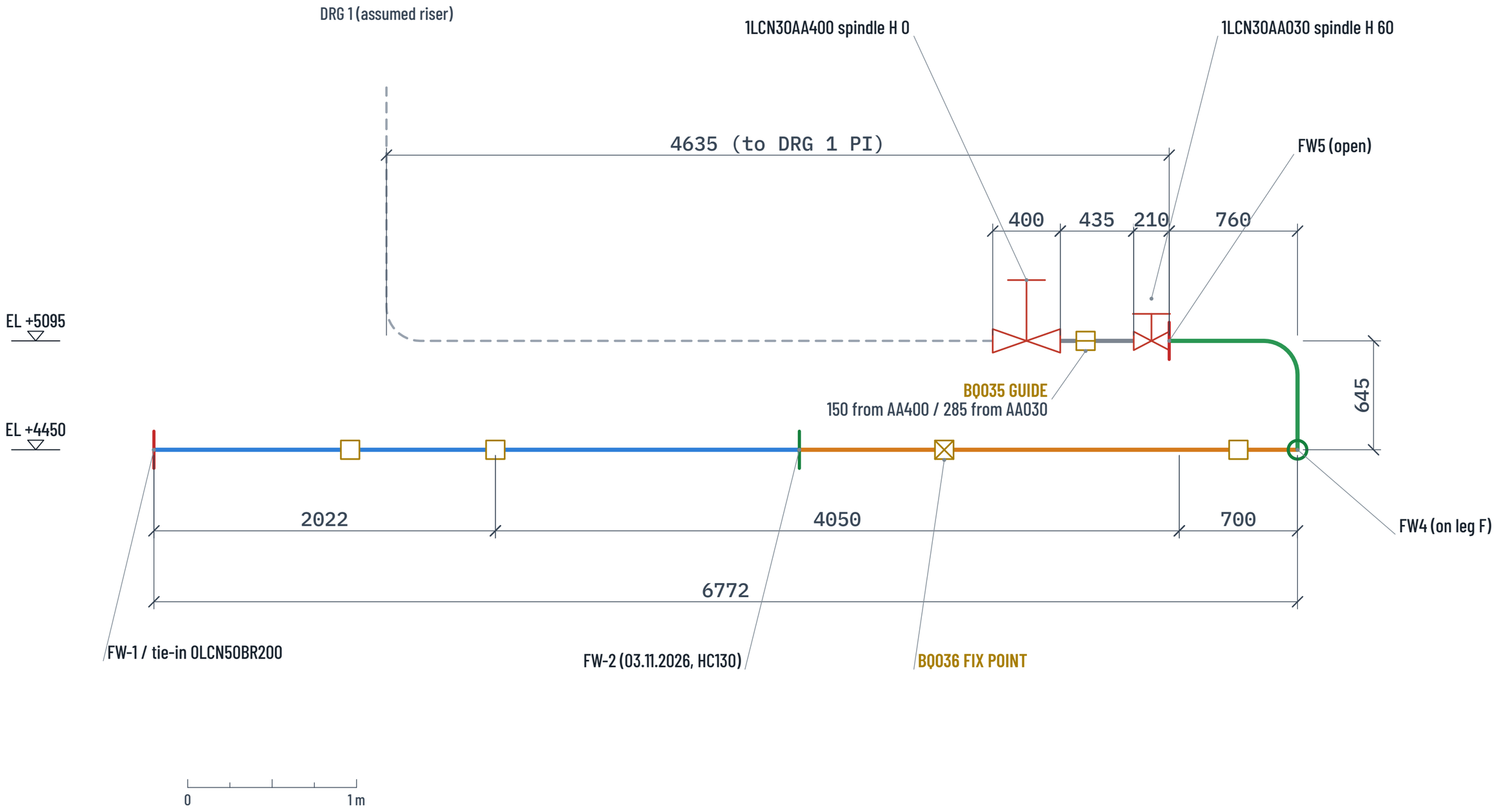


NOTE: valve run (AA400, item <8>, AA030, leg H) is at EL +5095, directly above segment C.

1LCN30 · DRG 2
DN40 condensate · class ECJB_TBY · P&ID 90207086

ELEVATION A - looking North (from South)
X horizontal (East to the right), Z vertical - true scale

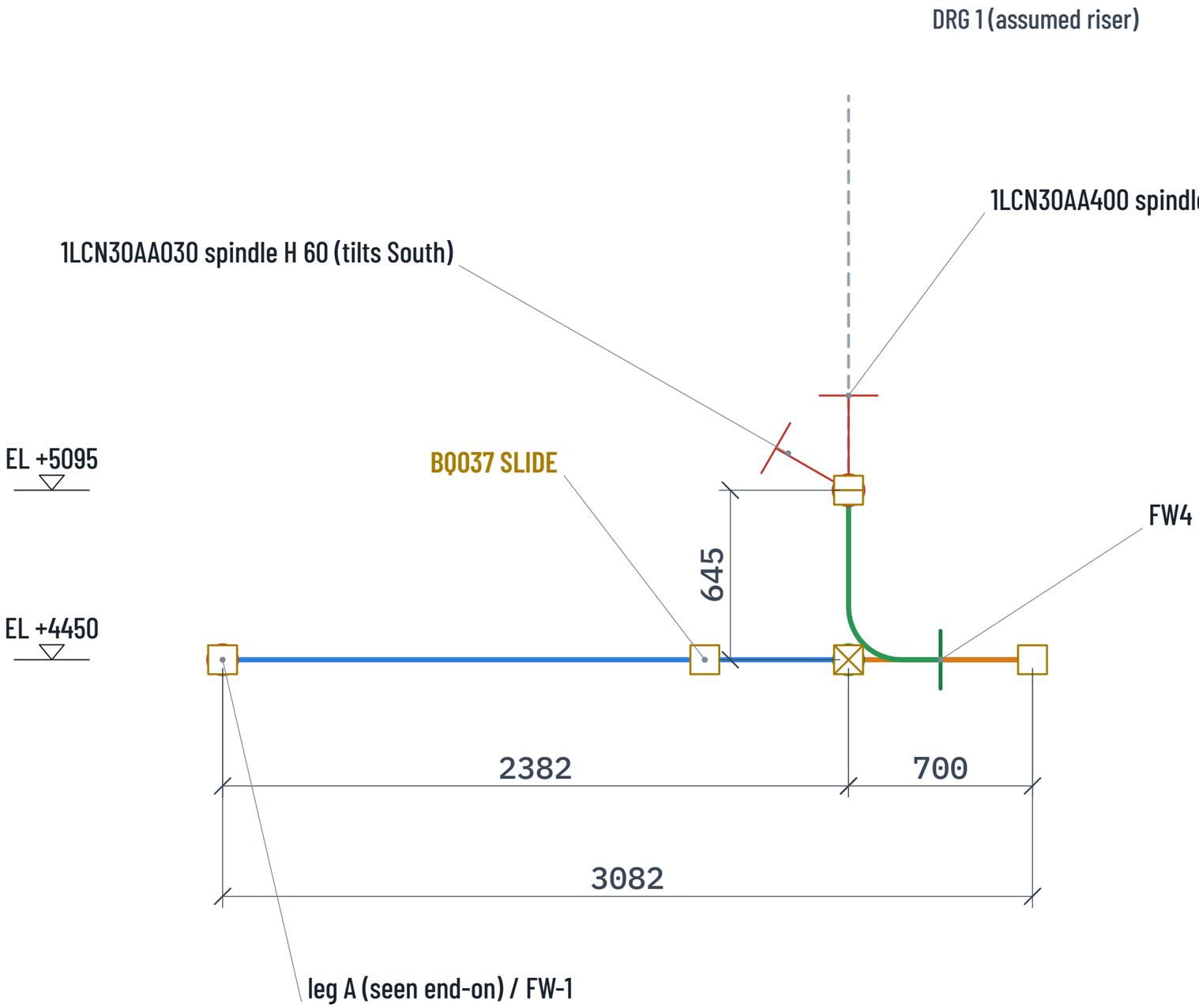
06.11.2026 · reconstructed from site iso · not an issued drawing



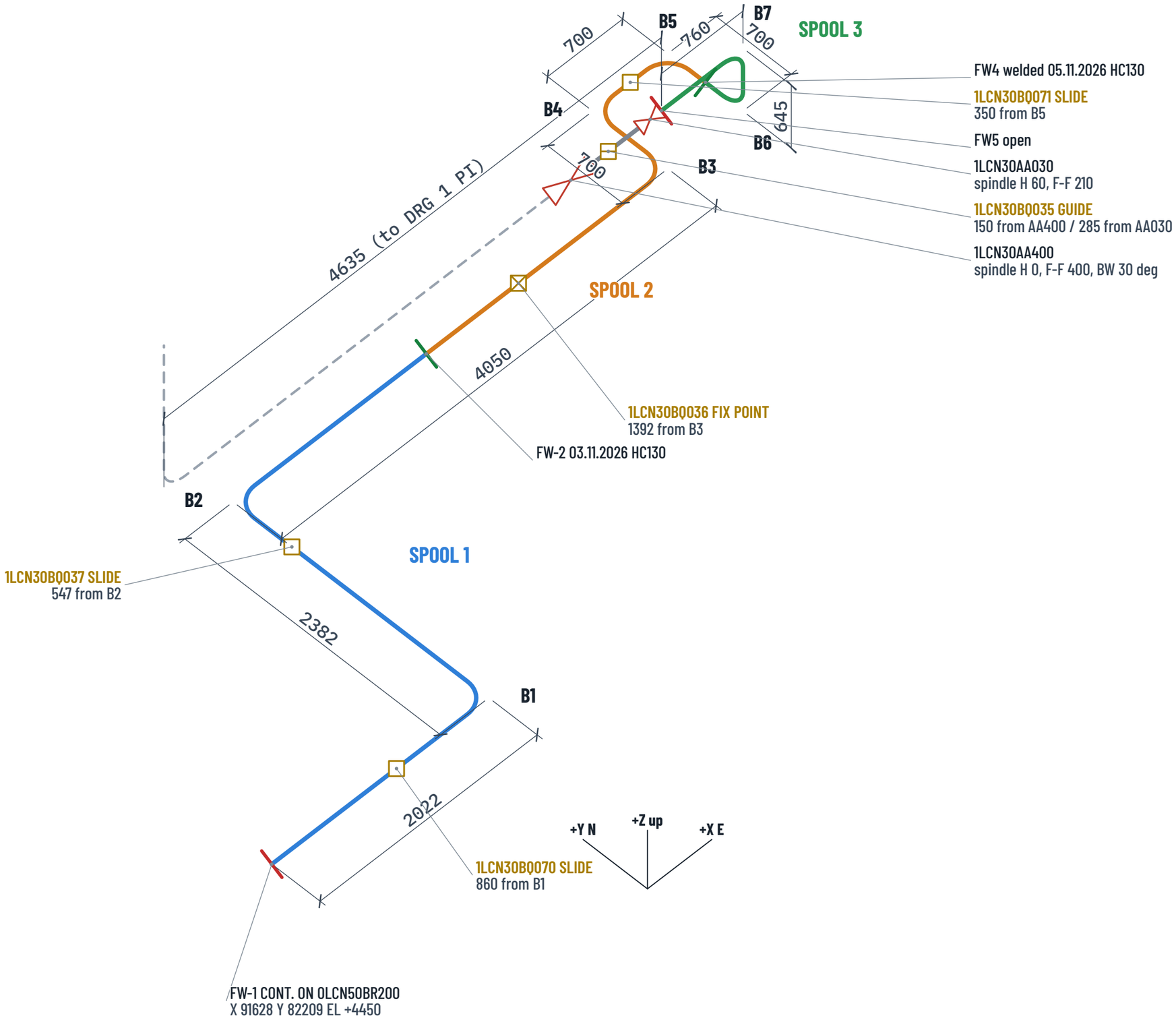
1LCN30 · DRG 2
DN40 condensate · class ECJB_TBY · P&ID 90207086

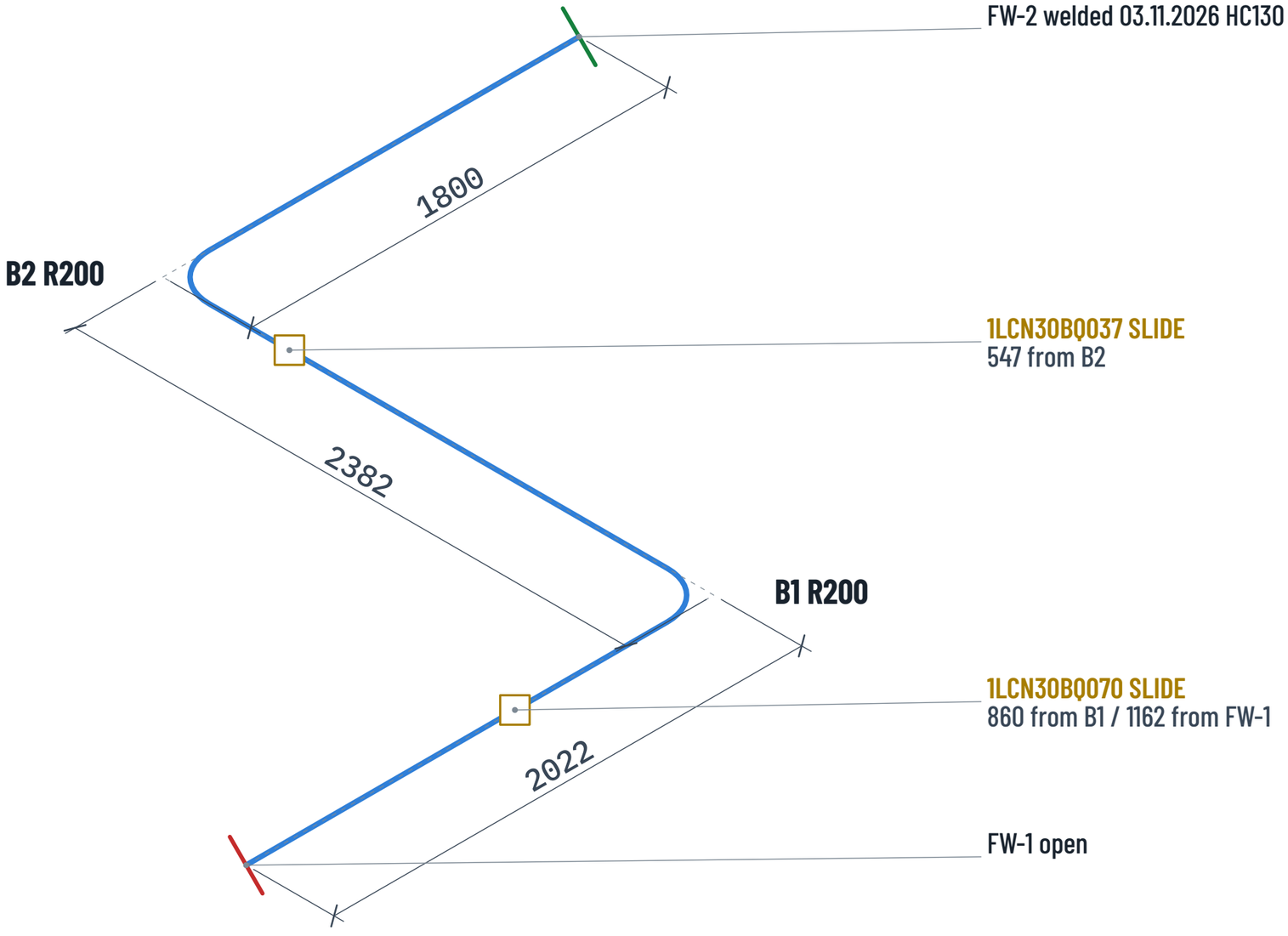
ELEVATION B - looking West (from East)
Y horizontal (North to the right), Z vertical - true scale

Sheet 5 of 12
06.11.2026 · reconstructed from site iso · not an issued drawing



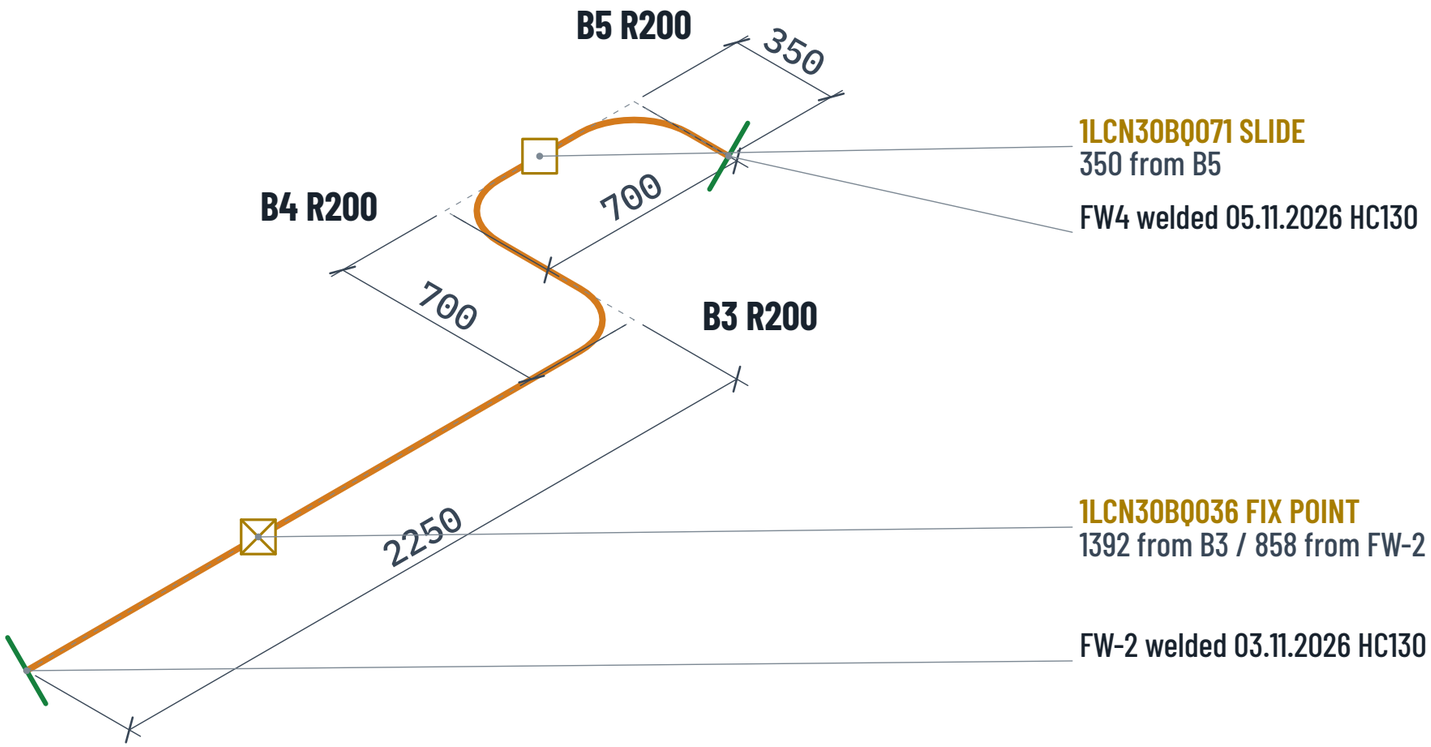
1LCN30 · DRG 2	3D VIEW - axonometric from South-West (true scale)	Sheet 6 of 12
DN40 condensate · class ECJB_TBY · P&ID 90207086	+X (E) up-right, +Y (N) up-left as on the isometric; 50 deg elevation so the EL +5095 run does not hide the loop	06.11.2026 · reconstructed from site iso · not an issued drawing





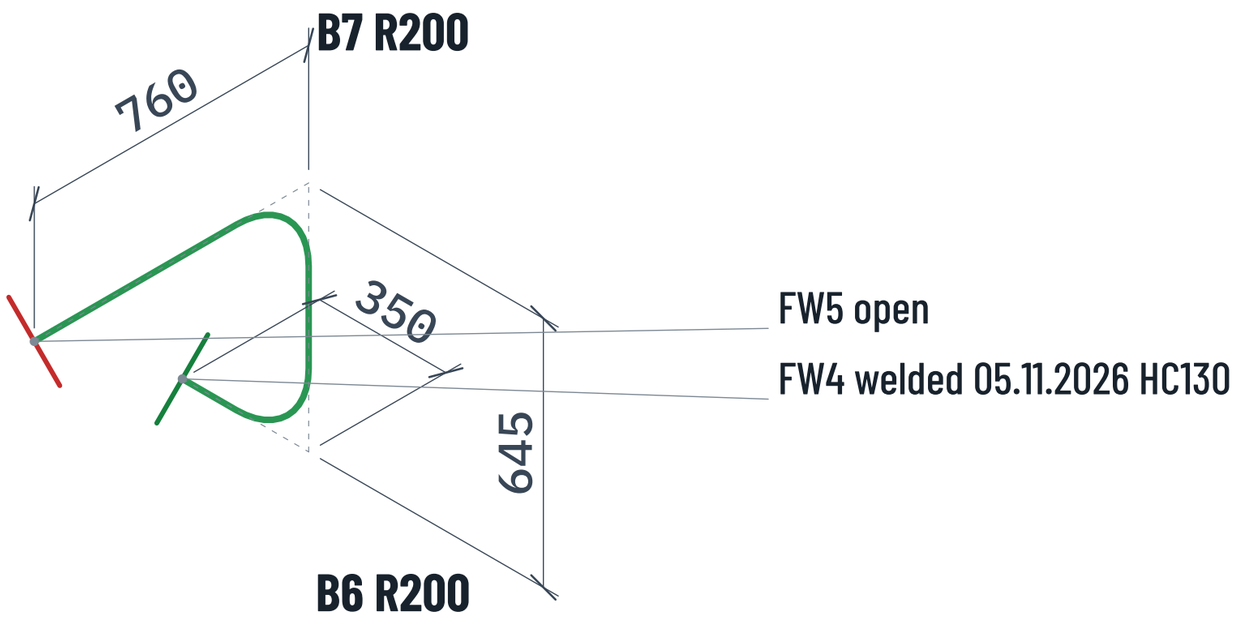
SPOOL 1 CUT DATA

Legs PI-PI: 2022 / 2382 / 1800
Directions: +X (E), +Y (N), +X (E)
Bends: 2 x 90 deg R200 (B1, B2)
Straights: 1822 / 1982 / 1600
Arc length per bend: 314.2
Developed cut length: 6032 mm
Bending (Y-B-C):
Y 1822 B +0 C 90 (B1)
Y 1982 B +180 C 90 (B2)
Y 1600 (to FW-2)
Add site allowance on field-fit end
(per iso general remarks)



SPOOL 2 CUT DATA

Legs PI-PI: 2250 / 700 / 700 / 350
Directions: +X (E), +Y (N), +X (E), -Y (S)
Bends: 3 x 90 deg R200 (B3, B4, B5)
Straights: 2050 / 300 / 300 / 150
Arc length per bend: 314.2
Developed cut length: 3742 mm
Bending (Y-B-C):
Y 2050 B +0 C 90 (B3)
Y 300 B +180 C 90 (B4)
Y 300 B +0 C 90 (B5)
Y 150 (to FW4)
Add site allowance on field-fit end
(per iso general remarks)



SPOOL 3 CUT DATA

Legs PI-PI: 350 / 645 / 760
Directions: -Y (S), +Z (up), -X (W)
Bends: 2 x 90 deg R200 (B6, B7)
Straights: 150 / 245 / 560
Arc length per bend: 314.2
Developed cut length: 1583 mm
Bending (Y-B-C):
Y 150 B +0 C 90 (B6)
Y 245 B +90 C 90 (B7)
Y 560 (to FW5)
Add site allowance on field-fit end
(per iso general remarks)

Spool 1 · FW-1 → FW-2

6032.3 mm developed cut length

LEG	PI-PI	STRAIGHT	DIRECTION
A	2 022	1 822	+X(E)
B	2 382	1 982	+Y(N)
C	1 800	1 600	+X(E)

Bends B1, B2: 90°, R 200, arc 314.2 each. PI sum 6 204.

`Y 1 822 / B +0° / C 90° · Y 1 982 / B +180° / C 90° · Y 1 600`

From E 91628 / N 82209 / EL +4450
to E 95450 / N 84591 / EL +4450

Spool 2 · FW-2 → FW4

3742.5 mm developed cut length

LEG	PI-PI	STRAIGHT	DIRECTION
A	2 250	2 050	+X(E)
B	700	300	+Y(N)
C	700	300	+X(E)
D	350	150	-Y(S)

Bends B3, B4, B5: 90°, R 200, arc 314.2 each. PI sum 4 000.

`Y 2 050 / B +0° / C 90° · Y 300 / B +180° / C 90° · Y 300 / B +0° / C 90° · Y 150`

From E 95450 / N 84591 / EL +4450
to E 98400 / N 84941 / EL +4450

Spool 3 · FW4 → FW5

1583.3 mm developed cut length

LEG	PI-PI	STRAIGHT	DIRECTION
A	350	150	-Y(S)
B	645	245	+Z(up)
C	760	560	-X(W)

Bends B6, B7: 90°, R 200, arc 314.2 each. PI sum 1 755.

`Y 150 / B +0° / C 90° · Y 245 / B +90° / C 90° · Y 560`

From E 98400 / N 84941 / EL +4450
to E 97640 / N 84591 / EL +5095

Bend tangent points (plant coordinates X / Y / EL)

BEND	ANGLE	TANGENT	TP1	TP2	CENTRE
B1	90.0°	200.0	93 450 / 82 209 / 4 450	93 650 / 82 409 / 4 450	93 450 / 82 409 / 4 450
B2	90.0°	200.0	93 650 / 84 391 / 4 450	93 850 / 84 591 / 4 450	93 850 / 84 391 / 4 450
B3	90.0°	200.0	97 500 / 84 591 / 4 450	97 700 / 84 791 / 4 450	97 500 / 84 791 / 4 450
B4	90.0°	200.0	97 700 / 85 091 / 4 450	97 900 / 85 291 / 4 450	97 900 / 85 091 / 4 450
B5	90.0°	200.0	98 200 / 85 291 / 4 450	98 400 / 85 091 / 4 450	98 200 / 85 091 / 4 450
B6	90.0°	200.0	98 400 / 84 791 / 4 450	98 400 / 84 591 / 4 650	98 400 / 84 791 / 4 650
B7	90.0°	200.0	98 400 / 84 591 / 4 895	98 200 / 84 591 / 5 095	98 200 / 84 591 / 4 895

Route points

POINT	X (E)	Y (N)	EL	DESCRIPTION
FW-1	91 628	82 209	+4 450	Field weld / tie-in 0LCN50BR200
B1	93 650	82 209	+4 450	Bend 90 deg, R 200 (PI)
B2	93 650	84 591	+4 450	Bend 90 deg, R 200 (PI)
B3	97 700	84 591	+4 450	Bend 90 deg, R 200 (PI)
B4	97 700	85 291	+4 450	Bend 90 deg, R 200 (PI)
B5	98 400	85 291	+4 450	Bend 90 deg, R 200 (PI)
B6	98 400	84 591	+4 450	Bend 90 deg, R 200 (PI)
B7	98 400	84 591	+5 095	Bend 90 deg, R 200 (PI)
FW5	97 640	84 591	+5 095	Field weld to valve 1LCN30AA030
FW-2	95 450	84 591	+4 450	Field weld spool 1/2 (site)
FW4	98 400	84 941	+4 450	Field weld spool 2/3 (site)
AA030-U	97 430	84 591	+5 095	Valve AA030 upstream end
BQ035	97 145	84 591	+5 095	Guide support
AA400-D	96 995	84 591	+5 095	Valve AA400 downstream end
AA400-U	96 595	84 591	+5 095	Valve AA400 upstream end (CONT. FROM DRG 1)
DRG1-PI	93 005	84 591	+5 095	Upstream PI on DRG 1 (4635 from FW5)

Bill of material

#	DESCRIPTION	QTY	NOTE
1	Pipe DN40 (OD 48.3), class ECJB_TBY - spools 1-3 (developed)	11.36 m	wall per class ECJB_TBY
2	Pipe DN40 - piece between valves (iso ref <8>)	0.44 m	435 mm
3	Pipe bend 90 deg, R = 200 mm (bent on pipe)	7 pcs	B1-B7
4	Valve 1LCN30AA030 (iso item 2), F-F 210, spindle H 60 deg	1 pc	free issue / per valve list
5	Valve 1LCN30AA400 (iso item 3), F-F 400, spindle H 0 deg, 30 deg BW ends	1 pc	free issue / per valve list
6	Support S4 SLIDE (BQ070, BQ037, BQ071)	3 pcs	TC021-AFR-...-SU
7	Support S4/7/6/6/6 FIX POINT (BQ036)	1 pc	TC021-AFR-1LCN30BQ036-SU
8	Support S4/5/6/6/6 GUIDE (BQ035)	1 pc	TC021-AFR-1LCN30BQ035-SU
9	Field welds FW-1, FW-2, FW4, FW5 (+ valve welds VW-1..3)	4 + 3	FW-2, FW4 welded
10	Insulation 100 mm (density 0 per sheet)	11.8 m	excl. valves

Weld map

WELD	STATUS	DATE / WELDER	JOINT · LOCATION
FW-1	Open		Spool 1 to tie-in 0LCN50BR200 · Start of line, CONT. ON 0LCN50BR200
FW-2	Welded	03.11.2026 HC130	Spool 1 / Spool 2 · 1800 from B2 (site mark '1800+')
FW4	Welded	05.11.2026 HC130	Spool 2 / Spool 3 · ~350 from B5 (scaled from mark-up)
FW5	Open		Spool 3 to valve 1LCN30AA030 · 760 from B7
FW-3	Check		Not marked on this sheet · check DRG 1 / valve side
VW-1	Check		Valve AA030 / pipe item <8> · valve end (not numbered on sheet)
VW-2	Check		Pipe item <8> / valve AA400 · valve end (not numbered on sheet)
VW-3	Check		Valve AA400 / DRG 1 (30 deg bevel) · valve end (not numbered on sheet)

Supports

TAG	FUNCTION	LOCATION	X	Y	EL	DRAWING
1LCN30BQ070	SLIDE S4	860 from B1	92 790	82 209	+4 450	TC021-AFR-1LCN30BQ070-SU
1LCN30BQ037	SLIDE S4	547 from B2	93 650	84 044	+4 450	TC021-AFR-1LCN30BQ037-SU
1LCN30BQ036	FIX POINT S4/7/6/6/6	1392 from B3	96 308	84 591	+4 450	TC021-AFR-1LCN30BQ036-SU
1LCN30BQ071	SLIDE S4	350 from B5	98 050	85 291	+4 450	TC021-AFR-1LCN30BQ071-SU
1LCN30BQ035	GUIDE S4/5/6/6/6	285 from valve AA030 (on item <8>)	97 145	84 591	+5 095	TC021-AFR-1LCN30BQ035-SU

What was read and what was assumed

1. Orientation from the axis legend in the iso title block: +X drawn up-right, +Y up-left. Confirm with the north arrow on the issued iso; if it differs, rotate about FW-1.
2. Start point is the iso tie-in: X 91628 · Y 82209 · EL +4450 (0LCN50BR200).
3. Leg G rises 645, so the valve run is at EL +5095, directly above segment C.
4. FW-2 taken as 1800 from B2 from the site mark "1800+". FW4 scaled from the mark-up at about 350 from B5. Measure both on site.
5. FW-3 is not marked on this sheet. VW-1 to VW-3 (valve welds) are numbered here for reference only.
6. Valve and support geometry is symbolic. Real sizes: valve list and TC021-AFR-...-SU support drawings. Pipe OD 48.3 (DN40); wall per class ECJB_TBY.
7. Beyond valve AA400 the line continues on DRG 1 (4635 from FW5 to the next PI). The riser there is assumed from the 3D print.
8. Weld status from the hand mark-up: FW-2 welded 03.11.2026 and FW4 welded 05.11.2026, both welder HC130. FW-1 and FW5 have no record on the sheet.

Loading the model

1. **Navisworks Manage / Simulate:** open the plant model, Home › Append › 1LCN30_DRG2_DN40_3D_Model.dxf (units mm) or the .fbx. It lands at its real X / Y / EL.
2. **Navisworks Freedom** opens NWD only and cannot append. A Manage user appends and saves an NWD; Freedom then opens it.
3. **AutoCAD / Plant 3D / BricsCAD:** open the 3D DXF in model space (WCS = plant coordinates). Layers: PIPE-SPOOL1..3, PIPE-ITEM8, INSULATION, VALVES, SUPPORTS, WELDS-DONE / OPEN / VALVE, CENTERLINE, LABELS, DRG1-CONTEXT. The 2D DXF holds all seven sheets.
4. **Interactive viewer:** 1LCN30_DRG2_DN40_Viewer.html opens in any browser (internet needed for the 3D library): orbit, click for properties, measure, plant coordinates under the cursor.

Package

`…_Drawing_Set.pdf`	this document
`…_3D_Model.dxf / .fbx / .obj / .blend`	3D model, plant coordinates mm, Z up
`…_2D_Drawings.dxf`	seven 2D sheets
`…_Viewer.html`	interactive 3D / 2D viewer
`…_*.csv`	coordinates, spools, welds, supports, BOM
`renders/*.png`	3D images